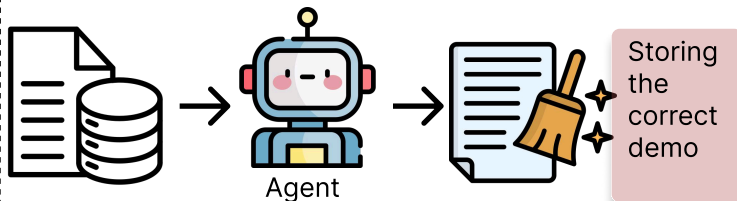




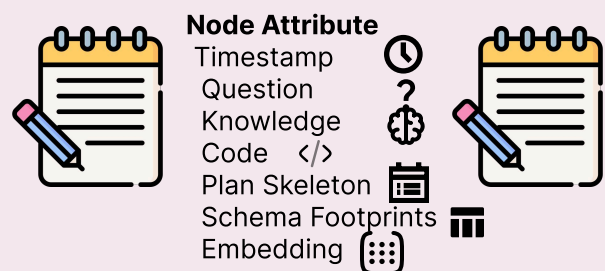
Question: Count the number of patients who were prescribed heparin flush hickman (100 units/ml) since 2101.

Step 1: Demonstration Mining

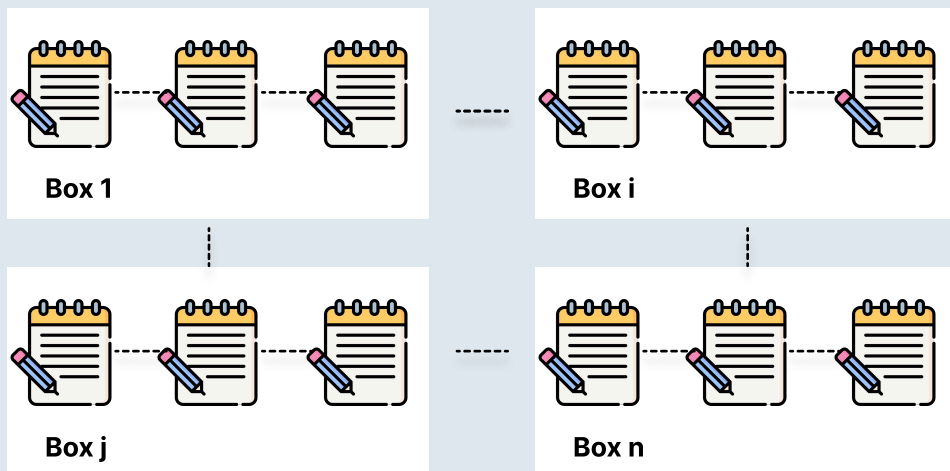


Step 2: Graph Memory Building

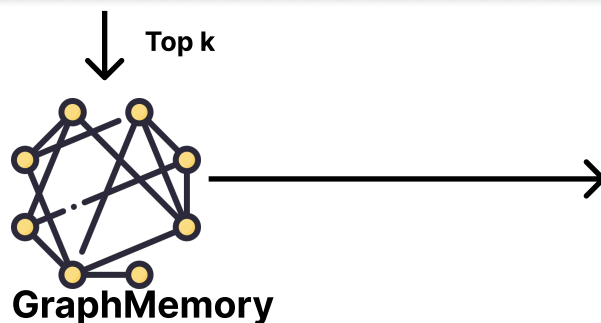
Node Construction (V)



Link Generation (E)



Each node connects to its top-K most similar nodes based on embedding similarity. These links collectively form the memory graph $G=(V,E)$



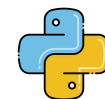
Step 4: Code Generation with Execution Feedback

```
SELECT COUNT(DISTINCT
p.SUBJECT_ID)
FROM prescriptions p
JOIN d_items d ON p.ITEMID =
d.ITEMID
WHERE d.LABEL = 'heparin flush
hickman (100 units/ml)'
AND p.STARTDATE >= '2101-01-01';
```



Tool Set

SQLInterpreter(q)
LoadDB(name)
FilterDB(conditions)
GetValue(field)



Executor

Question,
Code,
Error
message

Runtime Error: no such column:
p.ITEMID



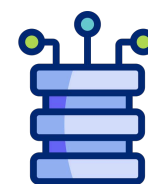
Debugger

Potential Reasons:
**Single most likely
reason for the error:**

```
SELECT COUNT(DISTINCT
SUBJECT_ID)
FROM prescriptions
WHERE DRUG = 'heparin flush
hickman (100 units/ml)'
AND STARTDATE >= '2101-01-01';
```

Output: 2101-01-01

- Medication prescription records are stored in the prescriptions table
- Dictionary tables such as d_items are typically joined with event tables
- Temporal constraints should be applied using the prescription start date



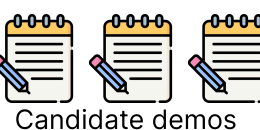
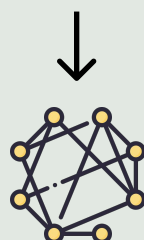
Contextual
EHR
Knowledge

Updated k-shot
demos



Step 3: RL Demo Retrieval

Question



Candidate demos

Top-k demonstrations

RL policy π_θ
score(d) = $\phi(d)^T \theta$